

COURSE CODE

SBT-DF202

COURSE TITLE

Computer and Digital Forensics

LAB TITLE

Data Carving with XXD, Binwalk, and Scalpel

NAME

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STUDENT ID

2025/FWSD/11325

DATE

05/09/2026

PLATFORM USED

Kali Linux

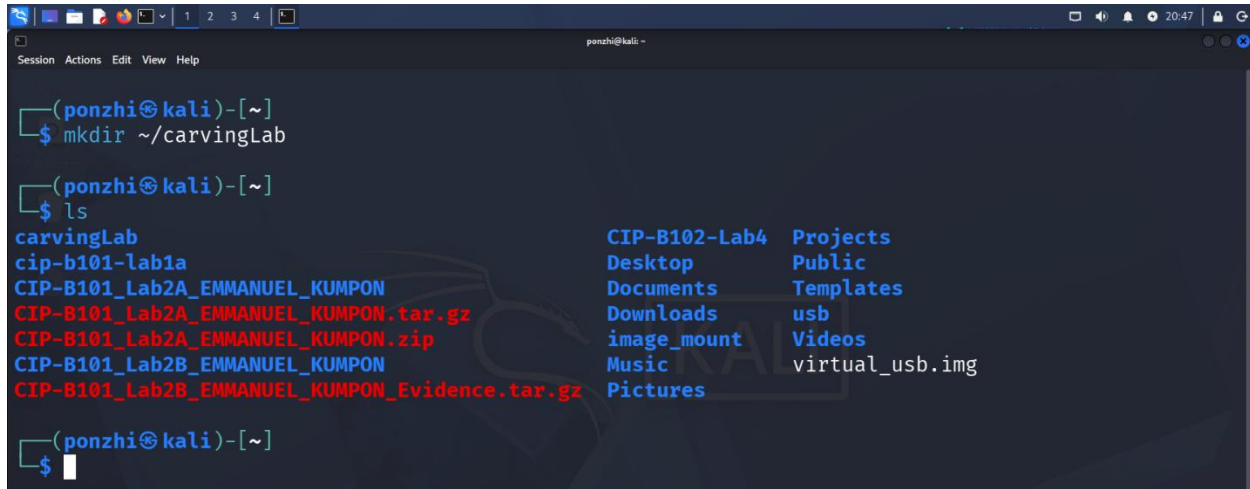
INSTRUCTOR

Mr. Aminu Idris, AMCPN
Commandant

OBJECTIVES

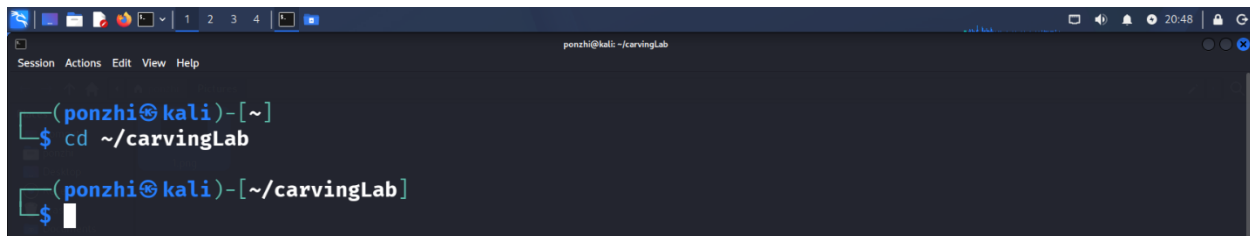
By completing this lab, I have demonstrate the ability to explain why data carving is required when file-system metadata is missing or unreliable, using xxd to examine binary file content and identify headers and footers, recognize JPEG SOI and EOI signatures, reconstruct binary content from a hexadecimal dump, using the Sleuth Kit to examine inode, block and sector-level data, identify and extract embedded content using Binwalk; recover deleted files from a forensic USB image using Scalpel, verify recovered evidence using cryptographic hashes; and prepare a professional forensic report containing commands, screenshots, findings and limitations.

Creating a dedicated working directory



```
(ponzhi@kali)-[~]
$ mkdir ~/carvingLab

(ponzhi@kali)-[~]
$ ls
carvingLab          CIP-B102-Lab4  Projects
cip-b101-lab1a      Desktop        Public
CIP-B101_Lab2A_EMMANUEL_KUMPON  Documents      Templates
CIP-B101_Lab2A_EMMANUEL_KUMPON.tar.gz  Downloads      usb
CIP-B101_Lab2A_EMMANUEL_KUMPON.zip      image_mount    Videos
CIP-B101_Lab2B_EMMANUEL_KUMPON          Music           virtual_usb.img
CIP-B101_Lab2B_EMMANUEL_KUMPON_Evidence.tar.gz  Pictures
```

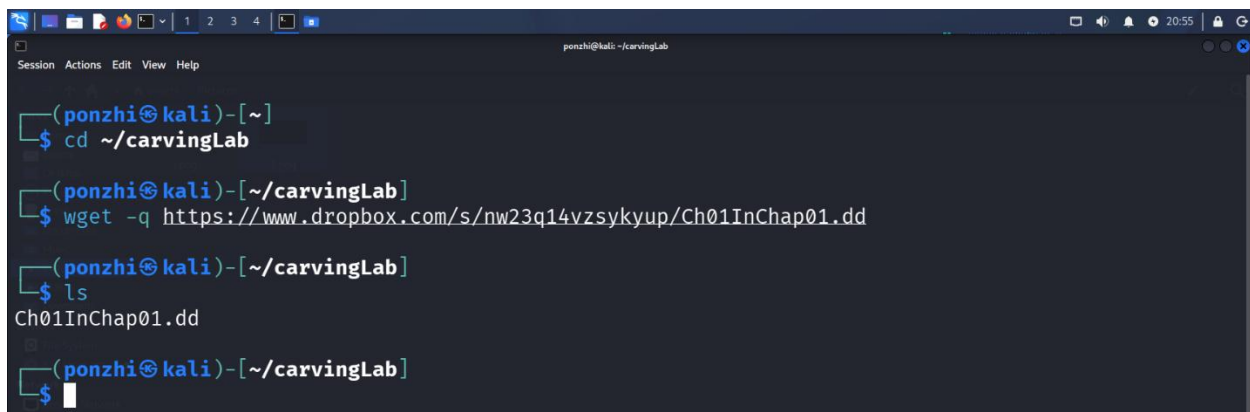


```
(ponzhi@kali)-[~]
$ cd ~/carvingLab

(ponzhi@kali)-[~/carvingLab]
$
```

Downloading evidence and hashes

Ch01InChap01.dd



```
(ponzhi@kali)-[~]
$ cd ~/carvingLab

(ponzhi@kali)-[~/carvingLab]
$ wget -q https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd

(ponzhi@kali)-[~/carvingLab]
$ ls
Ch01InChap01.dd

(ponzhi@kali)-[~/carvingLab]
$
```

J_ub_law.jpg and 120M.7z

```
ponzhi@kali: ~/carvingLab
Session Actions Edit View Help

(ponzhi@kali)~[~/carvingLab]
$ wget -q https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd

(ponzhi@kali)~[~/carvingLab]
$ ls
Ch01InChap01.dd

(ponzhi@kali)~[~/carvingLab]
$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_file/J_ub_law.jpg

(ponzhi@kali)~[~/carvingLab]
$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z

(ponzhi@kali)~[~/carvingLab]
$ ls
120M.7z Ch01InChap01.dd J_ub_law.jpg

(ponzhi@kali)~[~/carvingLab]
$
```

```
ponzhi@kali: ~/carvingLab
Session Actions Edit View Help

(ponzhi@kali)~[~/carvingLab]
$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_file/J_ub_law.jpg

(ponzhi@kali)~[~/carvingLab]
$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z

(ponzhi@kali)~[~/carvingLab]
$ ls
120M.7z Ch01InChap01.dd J_ub_law.jpg

(ponzhi@kali)~[~/carvingLab]
$ ls -l
total 38956
-rw-rw-r-- 1 ponzhi ponzhi 36720470 Sep  2 21:05 120M.7z
-rw-rw-r-- 1 ponzhi ponzhi 1474560 Sep  2 20:55 Ch01InChap01.dd
-rw-rw-r-- 1 ponzhi ponzhi 1692150 Sep  2 20:56 J_ub_law.jpg

(ponzhi@kali)~[~/carvingLab]
$
```

Hashes

```
ponzhi@kali: ~/carvingLab
$ hashdeep -c md5,sha256 Ch01InChap01.dd J_ub_law.jpg 120M.7z | tee evidence_hashes_original.txt
HASHDEEP-1.0
size,md5,sha256,filename
Invoked from: /home/ponzhi/carvingLab
$ hashdeep -c md5,sha256 Ch01InChap01.dd J_ub_law.jpg 120M.7z
1474560,a117773bcf1fc88ec0ab8e0a349fbbcb,3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122,/home/
ponzhi/carvingLab/Ch01InChap01.dd
1692150,83a360ac7f7e0ca318e5bfe39f95f137,238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468,/home/
ponzhi/carvingLab/J_ub_law.jpg
36720470,dfe7b5424e54cd1bf50d5df47aceeb3c,2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce,/home/
ponzhi/carvingLab/120M.7z
ponzhi@kali:~/carvingLab
$
```

Part A - File Signatures and Hex Analysis

Confirm the file type and look at the header/tail with xxd

```
ponzhi@kali: ~/carvingLab
$ file J_ub_law.jpg
J_ub_law.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height
=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper
-left], baseline, precision 8, 1920x1080, components 3
ponzhi@kali:~/carvingLab
$ xxd J_ub_law.jpg | head
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201 .....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100 .....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400 .....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201 .....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100 .....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00 .....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801 .....(.
00000090: 0300 0100 0000 0200 0000 3101 0200 2100 .....1...!.
ponzhi@kali:~/carvingLab
$
```

```
ponzhi@kali: ~/carvingLab
Session Actions Edit View Help
J_ub_law.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3

(ponzhi@kali)~[~/carvingLab]
$ xxd J_ub_law.jpg | head
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....0.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201 .....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100 .....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400 .....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201 .....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100 .....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00 .....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801 .....(
00000090: 0300 0100 0000 0200 0000 3101 0200 2100 .....1...!.

(ponzhi@kali)~[~/carvingLab]
$ xxd J_ub_law.jpg | tail -n 3
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1 ..*-$....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565 ..RT.8..o..Z.T.e
0019d1f0: d351 9ebf ffd9 ..Q....

(ponzhi@kali)~[~/carvingLab]
$
```

JPEG files begin with ffd8 (SOI) and end with ffd9 (EOI).

Create a plain (unformatted) hex dump and reconstruct the file from it

```
ponzhi@kali: ~/carvingLab
Session Actions Edit View Help

(ponzhi@kali)~[~/carvingLab]
$ xxd -p J_ub_law.jpg > J_ub_law_hexdump.txt

(ponzhi@kali)~[~/carvingLab]
$ cat J_ub_law_hexdump.txt
ffd8ffe117f045786966000049492a000800000010000001030001000000
401300000101030001000000d00c00000201030003000000ce000000601
030001000000020000000f01020012000000d40000001001020009000000
e6000000120103000100000001501030001000000030000001a01
050001000000ef0000001b01050001000000f70000002801030001000000
020000003101020021000000ff0000003201020014000000200100009882
02000c000000340100009e9c010052000000400100006987040001000000
94010000440400000800080008004e494b4f4e20434f52504f524154494f
4e004e494b4f4e20443400c0c62d0010270000c0c62d001027000041646f
62652050686f746f73686f702032312e3120284d6163696e746f73682900
323032303a30383a32302031303a32303a303500484f57415244204b4f52
4e0032003000310033002c002000630061006d007000750073002c002000
4d00650072007200690063006b0020005300630068006f006f006c002000
6f006600200042007500730069006e006500730073000000000029009a82
050001000000860300009d820500010000008e0300002288030001000000
010000002788030001000000c8000000308803000100000002000000090
07000400000030323130390020014000000960300000490020014000000
aa03000001920a0001000000be0300000292050001000000c60300000492
0a0001000000ce0300000592050001000000d60300000792030001000000
```

Convert a plain hexdump to back into binary

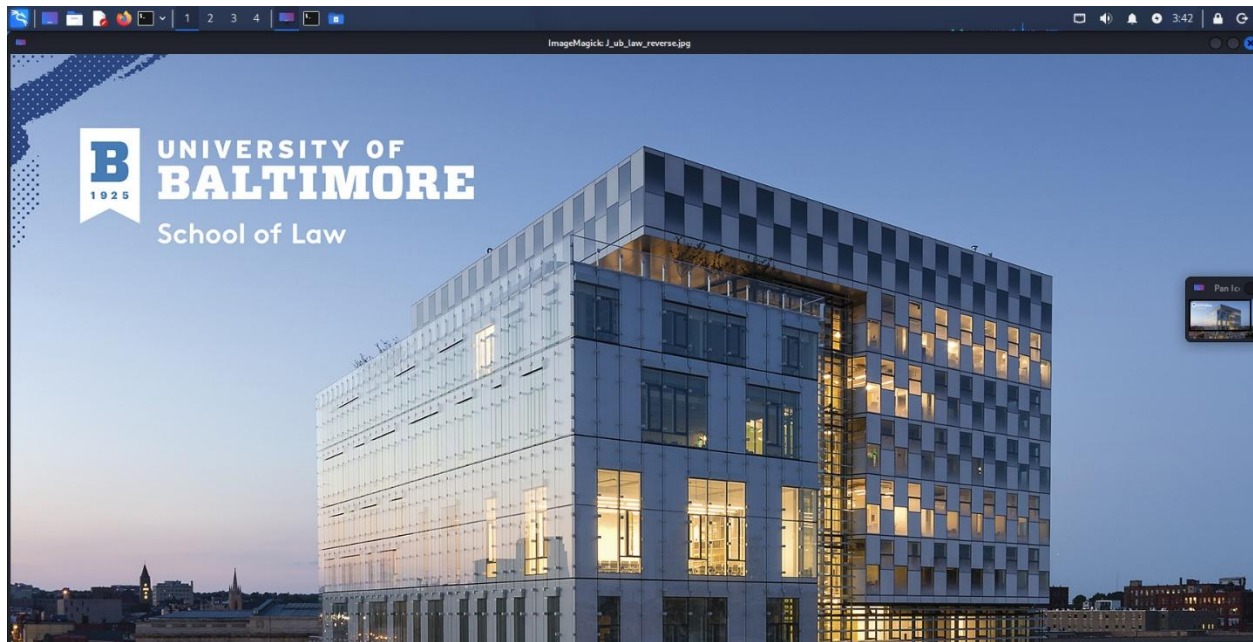
```
ponzhi@kali: ~/carvingLab
Session Actions Edit View Help

(ponzhi@kali)~[~/carvingLab]
$ xxd -r -p J_ub_law_hexdump.txt J_ub_law_reverse.jpg

(ponzhi@kali)~[~/carvingLab]
$ ls -l J_ub_law.jpg J_ub_law_reverse.jpg
-rw-rw-r-- 1 ponzhi ponzhi 1692150 Sep  2 20:56 J_ub_law.jpg
-rw-rw-r-- 1 ponzhi ponzhi 1692150 Sep  3 03:41 J_ub_law_reverse.jpg

(ponzhi@kali)~[~/carvingLab]
$ display J_ub_law_reverse.jpg
```

Image recovered



Verify the reconstruction with hashes

```
ponzhi@kali: ~/carvingLab
Session Actions Edit View Help

(ponzhi@kali)~[~/carvingLab]
$ md5sum J_ub_law.jpg J_ub_law_reverse.jpg
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law_reverse.jpg

(ponzhi@kali)~[~/carvingLab]
$ sha256sum J_ub_law.jpg J_ub_law_reverse.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law_reverse.jpg

(ponzhi@kali)~[~/carvingLab]
$
```

The hashes for the original and the reconstructed file match exactly.

Part B - Examine Embedded Metadata-Like Content

Broad text scan

```
ponzhi@kali: ~/carvingLab
$ strings J_ub_law.jpg | less
```

```
ponzhi@kali: ~/carvingLab
$ xxd J_ub_law.jpg | head -26
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201 .....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100 .....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400 .....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201 .....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100 .....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00 .....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801 .....(
00000090: 0300 0100 0000 0200 0000 3101 0200 2100 .....1...!.
000000a0: 0000 ff00 0000 3201 0200 1400 0000 2001 .....2.....
000000b0: 0000 9882 0200 0c00 0000 3401 0000 9e9c .....4....
000000c0: 0100 5200 0000 4001 0000 6987 0400 0100 ..R...@...i....
000000d0: 0000 9401 0000 4404 0000 0800 0800 0800 .....D.....
000000e0: 4e49 4b4f 4e20 434f 5250 4f52 4154 494f NIKON CORPORATIO
000000f0: 4e00 4e49 4b4f 4e20 4434 00c0 c62d 0010 N.NIKON D4...-..
00000100: 2700 00c0 c62d 0010 2700 0041 646f 6265 '....-..'..Adobe
00000110: 2050 686f 746f 7368 6f70 2032 312e 3120 Photoshop 21.1
00000120: 284d 6163 696e 746f 7368 2900 3230 3230 (Macintosh).2020
00000130: 3a30 383a 3230 2031 303a 3230 3a30 3500 :08:20 10:20:05.
00000140: 484f 5741 5244 204b 4f52 4e00 3200 3000 HOWARD KORN.2.0.
00000150: 3100 3300 2c00 2000 6300 6100 6d00 7000 1.3.,. .c.a.m.p.
```

```
ponzhi@kali: ~/carvingLab
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400 .....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201 .....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100 .....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00 .....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801 .....(
00000090: 0300 0100 0000 0200 0000 3101 0200 2100 .....1...!.
000000a0: 0000 ff00 0000 3201 0200 1400 0000 2001 .....2.....
000000b0: 0000 9882 0200 0c00 0000 3401 0000 9e9c .....4....
000000c0: 0100 5200 0000 4001 0000 6987 0400 0100 ..R...@...i....
000000d0: 0000 9401 0000 4404 0000 0800 0800 0800 .....D.....
000000e0: 4e49 4b4f 4e20 434f 5250 4f52 4154 494f NIKON CORPORATIO
000000f0: 4e00 4e49 4b4f 4e20 4434 00c0 c62d 0010 N.NIKON D4...-..
00000100: 2700 00c0 c62d 0010 2700 0041 646f 6265 '....-..'..Adobe
00000110: 2050 686f 746f 7368 6f70 2032 312e 3120 Photoshop 21.1
00000120: 284d 6163 696e 746f 7368 2900 3230 3230 (Macintosh).2020
00000130: 3a30 383a 3230 2031 303a 3230 3a30 3500 :08:20 10:20:05.
00000140: 484f 5741 5244 204b 4f52 4e00 3200 3000 HOWARD KORN.2.0.
00000150: 3100 3300 2c00 2000 6300 6100 6d00 7000 1.3.,. .c.a.m.p.
00000160: 7500 7300 2c00 2000 4d00 6500 7200 7200 u.s.,. .M.e.r.r.
00000170: 6900 6300 6b00 2000 5300 6300 6800 6f00 i.c.k. .S.c.h.o.
00000180: 6f00 6c00 2000 6f00 6600 2000 4200 7500 o.l. .o.f. .B.u.
00000190: 7300 6900 6e00 6500 7300 7300 0000 0000 s.i.n.e.s.s....

ponzhi@kali: ~/carvingLab
$
```

Look through this for camera make/model, software, dates, or names

```
ponzhi@kali: ~/carvingLab
Session Actions Edit View Help
Exif
NIKON CORPORATION
NIKON D4
Adobe Photoshop 21.1 (Macintosh)
2020:08:20 10:20:05
HOWARD KORN
0221
2013:10:08 18:56:21
2013:10:08 18:56:21
2027750
17.0-35.0 mm f/2.8
Adobe_CM
Adobe
b34r
7GWgw
AQaq"
dEU6te
'7GWgw
2VtWX
I;[h
[$Wik
S_b[
Suhf
kC{ LO
:|
```

```
ponzhi@kali: ~/carvingLab
Session Actions Edit View Help
Merrick School of Business
HOWARD KORN
8BIM
8BIM
printOutput
ClrSenum
ClrS
RGBC
Inteenum
Inte
Clrm
MpBlbool
printSixteenBitbool
printerNameTEXT
printProofSetupObjc
proofSetup
Bltnenum
builtinProof
    proofCMYK
8BIM
printOutputOptions
Cptnbool
Clbrbool
RgsMbool
:|
```


Hexdump Photoshop info starts from 0x0e0 and list 0xbf bytes

```
ponzhi@kali: ~/carvingLab
$ xxd -s 0x0e0 -l 0xbf J_ub_low.jpg
000000e0: 4e49 4b4f 4e20 434f 5250 4f52 4154 494f  NIKON CORPORATIO
000000f0: 4e00 4e49 4b4f 4e20 4434 00c0 c62d 0010  N.NIKON D4...-..
00000100: 2700 00c0 c62d 0010 2700 0041 646f 6265  '...-..'..Adobe
00000110: 2050 686f 746f 7368 6f70 2032 312e 3120  Photoshop 21.1
00000120: 284d 6163 696e 746f 7368 2900 3230 3230  (Macintosh).2020
00000130: 3a30 383a 3230 2031 303a 3230 3a30 3500  :08:20 10:20:05.
00000140: 484f 5741 5244 204b 4f52 4e00 3200 3000  HOWARD KORN.2.0.
00000150: 3100 3300 2c00 2000 6300 6100 6d00 7000  1.3.,. .C.a.m.p.
00000160: 7500 7300 2c00 2000 4d00 6500 7200 7200  u.s.,. .M.e.r.r.
00000170: 6900 6300 6b00 2000 5300 6300 6800 6f00  i.c.k. .S.c.h.o.
00000180: 6f00 6c00 2000 6f00 6600 2000 4200 7500  o.l. .o.f. .B.u.
00000190: 7300 6900 6e00 6500 7300 7300 0000 00  s.i.n.e.s....

ponzhi@kali: ~/carvingLab
$
```

Part C - Direct Data-Unit Examination with The Sleuth Kit

Using Ch01InChap01.dd

Basic image info

img_stat Ch01InChap01.dd

```
ponzhi@kali: ~/carvingLab
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION

Image Type: raw

Size in bytes: 1474560
Sector size: 512

ponzhi@kali: ~/carvingLab
$
```

Partition layout needed before file-system-level commands, in case the image has multiple partitions

mmls Ch01InChap01.dd

```
ponzhi@kali: ~/carvingLab
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION

Image Type: raw

Size in bytes: 1474560
Sector size: 512

ponzhi@kali: ~/carvingLab
$ mmls Ch01InChap01.dd

ponzhi@kali: ~/carvingLab
$
```

File-system details use the correct offset from mmcls if the image isn't partition-less; omit -o if img_stat/mmcls shows a single filesystem starting at sector 0

fsstat Ch01InChap01.dd

```
ponzhi@kali: ~/carvingLab
$ fsstat Ch01InChap01.dd
FILE SYSTEM INFORMATION

File System Type: FAT12

OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: *3*4M|

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879
```

```
ponzhi@kali: ~/carvingLab
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION

Range: 2 - 45782
Root Directory: 2

CONTENT INFORMATION

Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

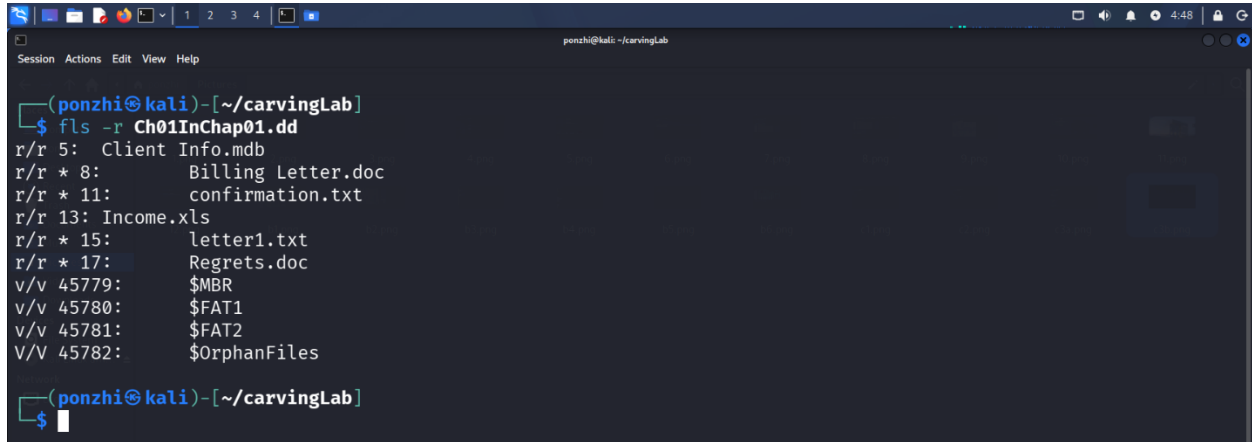
FAT CONTENTS (in sectors)

33-236 (204) → EOF
285-311 (27) → EOF

ponzhi@kali: ~/carvingLab
$
```

List files but allocated and deleted to find a real inode number to work with

fls -r Ch01InChap01.dd

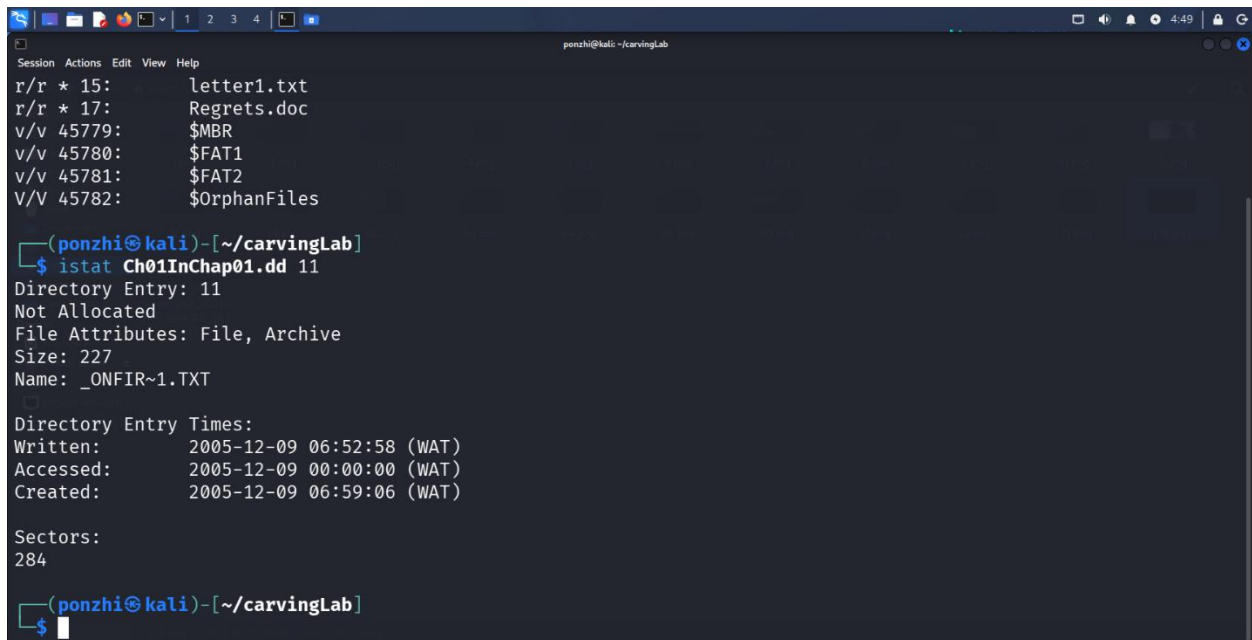


```
(ponzhi@kali)-[~/carvingLab]
$ fls -r Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles

(ponzhi@kali)-[~/carvingLab]
$
```

Get full metadata + the sector list for a specific file

istat Ch01InChap01.dd 11



```
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles

(ponzhi@kali)-[~/carvingLab]
$ istat Ch01InChap01.dd 11
Directory Entry: 11
Not Allocated
File Attributes: File, Archive
Size: 227
Name: _ONFIR~1.TXT

Directory Entry Times:
Written: 2005-12-09 06:52:58 (WAT)
Accessed: 2005-12-09 00:00:00 (WAT)
Created: 2005-12-09 06:59:06 (WAT)

Sectors:
284

(ponzhi@kali)-[~/carvingLab]
$
```

Extract that file's content by inode

icat Ch01InChap01.dd 11 > recovered_by_inode.out

```
ponzhi@kali: ~/carvingLab
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles

(ponzhi@kali)-[~/carvingLab]
$ istat Ch01InChap01.dd 11
Directory Entry: 11
Not Allocated
File Attributes: File, Archive
Size: 227
Name: _ONFIR~1.TXT

Directory Entry Times:
Written: 2005-12-09 06:52:58 (WAT)
Accessed: 2005-12-09 00:00:00 (WAT)
Created: 2005-12-09 06:59:06 (WAT)

Sectors:
284

(ponzhi@kali)-[~/carvingLab]
$ icat Ch01InChap01.dd 11 > recovered_by_inode.out

(ponzhi@kali)-[~/carvingLab]
$
```

Read specific sectors directly

blkcat Ch01InChap01.dd 284

```
ponzhi@kali: ~/carvingLab
$ blkcat Ch01InChap01.dd 284
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

login id: laura.roper
password: 342rroi9

Thank you for your business

George

(ponzhi@kali)-[~/carvingLab]
$
```

Part D - Embedded File Discovery with Binwalk

Since File_carving.docx was unavailable, Binwalk was run against J_ub_law.jpg instead, to fulfil the embedded-file-discovery objective still

```
ponzhi@kali: ~/carvingLab
$ binwalk File_carving.docx

General Error: Cannot open file File_carving.docx (CWD: /home/ponzhi/carvingLab) : [Errno 2] No such file or directory: 'File_carving.docx'

ponzhi@kali:~/carvingLab$
```

```
ponzhi@kali: ~/carvingLab
$ binwalk File_carving.docx

General Error: Cannot open file File_carving.docx (CWD: /home/ponzhi/carvingLab) : [Errno 2] No such file or directory: 'File_carving.docx'

ponzhi@kali:~/carvingLab$ binwalk J_ub_law.jpg

DECIMAL      HEXADECEMAL  DESCRIPTION
-----
0            0x0          JPEG image data, EXIF standard
12           0xC          TIFF image data, little-endian offset of first image directory: 8
26844       0x68DC       Copyright string: "Copyright 1999 Adobe Systems Incorporated"

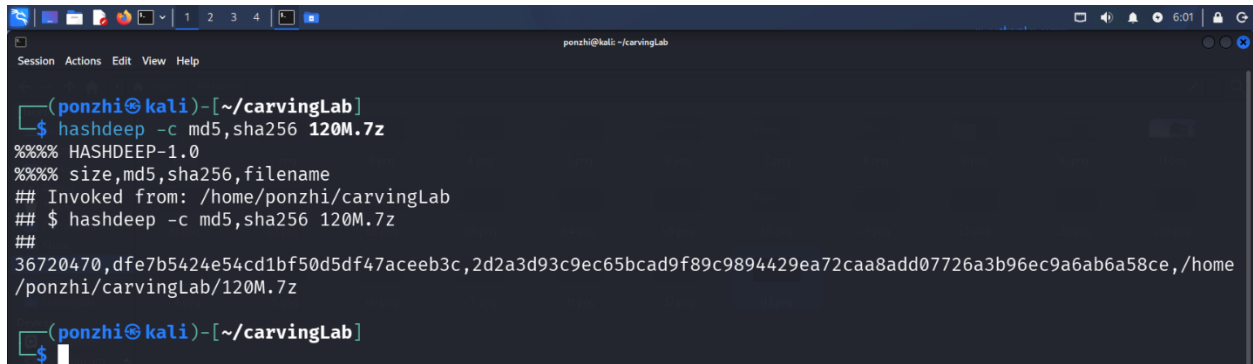
ponzhi@kali:~/carvingLab$
```

Part E - Prepare the USB Image

Using 120M.7z

Hash the archive as downloaded

```
hashdeep -c md5,sha256 120M.7z
```

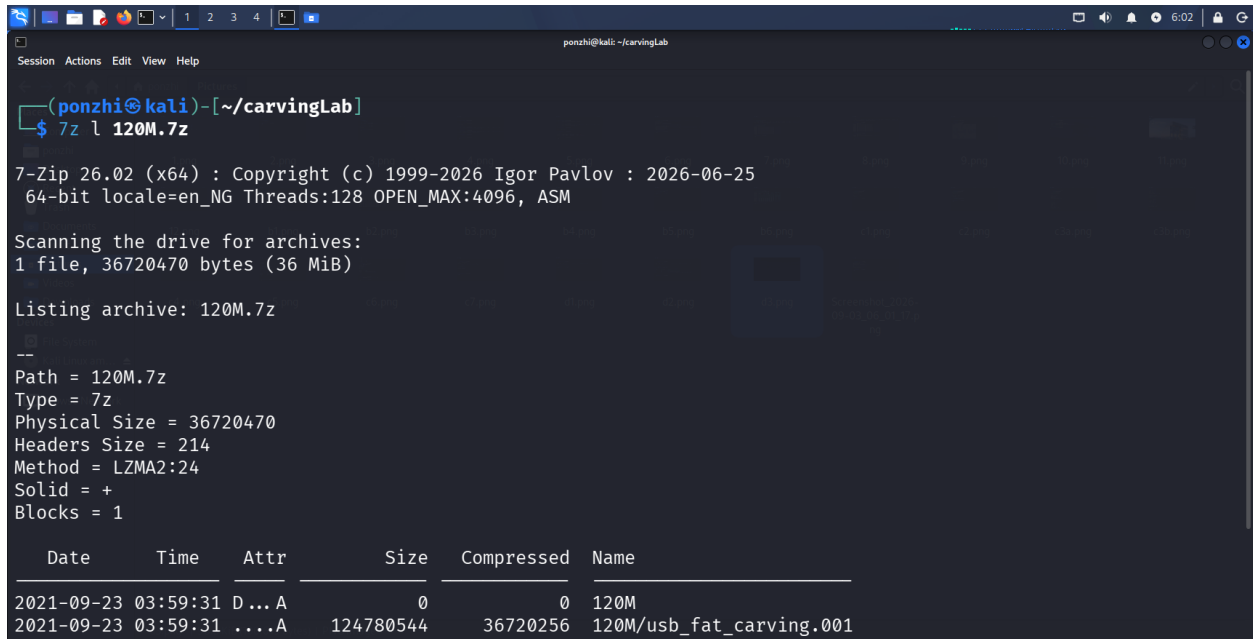


```
ponzhi@kali: ~/carvingLab
$ hashdeep -c md5,sha256 120M.7z
%%%% HASHDEEP-1.0
%%%% size,md5,sha256,filename
## Invoked from: /home/ponzhi/carvingLab
## $ hashdeep -c md5,sha256 120M.7z
##
36720470,dfe7b5424e54cd1bf50d5df47aceeb3c,2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce,/home/ponzhi/carvingLab/120M.7z

ponzhi@kali: ~/carvingLab
$
```

List contents before extracting

```
7z l 120M.7z
```



```
ponzhi@kali: ~/carvingLab
$ 7z l 120M.7z

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
64-bit locale=en_NG Threads:128 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Listing archive: 120M.7z
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

  Date       Time       Attr      Size    Compressed  Name
-----
2021-09-23  03:59:31  D...A         0          0    120M
2021-09-23  03:59:31  ....A   124780544   36720256  120M/usb_fat_carving.001
```



```
ponzhi@kali: ~/carvingLab
Session Actions Edit View Help

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Listing archive: 120M.7z
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

  Date       Time       Attr      Size    Compressed  Name
  ----
2021-09-23 03:59:31 D...A         0         0  120M
2021-09-23 03:59:31 ....A 124780544   36720256 120M/usb_fat_carving.001
2021-09-23 03:59:32 ....A    1685         0 120M/usb_fat_carving.001.txt
  ----
2021-09-23 03:59:32          124782229   36720256 2 files, 1 folders

(ponzhi@kali)-[~/carvingLab]
$
```

Extract and list the files

`7z e 120M.7z` `ls -l usb_fat_carving.001*`

```
ponzhi@kali: ~/carvingLab
Session Actions Edit View Help

(ponzhi@kali)-[~/carvingLab]
$ 7z e 120M.7z

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
64-bit locale=en_NG Threads:128 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Extracting archive: 120M.7z
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Folders: 1
Files: 2
Size:      124782229
Compressed: 36720470
```

```
Session Actions Edit View Help
ponzhi@kali: ~/carvingLab

Extracting archive: 120M.7z
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Folders: 1
Files: 2
Size:      124782229
Compressed: 36720470

(ponzhi@kali)-[~/carvingLab]
$ ls -l usb_fat_carving.001*
-rw-rw-r-- 1 ponzhi ponzhi 124780544 Sep 23  2021 usb_fat_carving.001
-rw-rw-r-- 1 ponzhi ponzhi    1685 Sep 23  2021 usb_fat_carving.001.txt

(ponzhi@kali)-[~/carvingLab]
$
```

Hash the extracted image and cross-check against the accompanying.

```
hashdeep -c md5,sha256 usb_fat_carving.001 cat usb_fat_carving.001.txt 2>/dev/null | grep -i checksum
```

```
Session Actions Edit View Help
ponzhi@kali: ~/carvingLab

(ponzhi@kali)-[~/carvingLab]
$ hashdeep -c md5,sha256 usb_fat_carving.001
%%%% HASHDEEP-1.0
%%%% size,md5,sha256,filename
## Invoked from: /home/ponzhi/carvingLab
## $ hashdeep -c md5,sha256 usb_fat_carving.001
##
124780544,ba4a1d0ba49f4a6667b00a3b3e85e604,9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9,/home/ponzhi/carvingLab/usb_fat_carving.001

(ponzhi@kali)-[~/carvingLab]
$ cat usb_fat_carving.001.txt 2>/dev/null | grep -i checksum
MD5 checksum: ba4a1d0ba49f4a6667b00a3b3e85e604
SHA1 checksum: bcc2d49fd49c9521ecb1739f6542c6bf327375ef
MD5 checksum: ba4a1d0ba49f4a6667b00a3b3e85e604 : verified
SHA1 checksum: bcc2d49fd49c9521ecb1739f6542c6bf327375ef : verified

(ponzhi@kali)-[~/carvingLab]
$
```

Confirm the partition structure — do NOT assume the offset is 128

fdisk -l usb_fat_carving.00, mmls usb_fat_carving.001

```
(ponzhi@kali)-[~/carvingLab]
$ fdisk -l usb_fat_carving.001
Disk usb_fat_carving.001: 119 MiB, 124780544 bytes, 243712 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa1159a00

Device                Boot Start    End Sectors  Size Id Type
usb_fat_carving.001p1 *      128 243711  243584 118.9M  e W95 FAT16 (LBA)

(ponzhi@kali)-[~/carvingLab]
$
```

```
(ponzhi@kali)-[~/carvingLab]
$ fdisk -l usb_fat_carving.001
Disk usb_fat_carving.001: 119 MiB, 124780544 bytes, 243712 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa1159a00

Device                Boot Start    End Sectors  Size Id Type
usb_fat_carving.001p1 *      128 243711  243584 118.9M  e W95 FAT16 (LBA)

(ponzhi@kali)-[~/carvingLab]
$ mmls usb_fat_carving.001
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

    Slot      Start      End      Length      Description
000:  Meta      0000000000  0000000000  0000000001  Primary Table (#0)
001:  _____ 0000000000  0000000127  0000000128  Unallocated
002:  000:000  0000000128  0000243711  0000243584  Win95 FAT16 (0x0e)

(ponzhi@kali)-[~/carvingLab]
$
```

Checking the file system at the confirmed offset

fsstat -o 128 usb_fat_carving.001

```
ponzhi@kali: ~/carvingLab
$ fsstat -o 128 usb_fat_carving.001
FILE SYSTEM INFORMATION
-----
File System Type: FAT16

OEM Name: MSDOS5.0
Volume ID: 0x94105672
Volume Label (Boot Sector): NO NAME
Volume Label (Root Directory): USB
File System Type Label: FAT16

Sectors before file system: 128

File System Layout (in sectors)
Total Range: 0 - 243583
* Reserved: 0 - 3
** Boot Sector: 0
* FAT 0: 4 - 241
* FAT 1: 242 - 479
* Data Area: 480 - 243583
** Root Directory: 480 - 511
** Cluster Area: 512 - 243583
```

```
ponzhi@kali: ~/carvingLab
* Data Area: 480 - 243583
** Root Directory: 480 - 511
** Cluster Area: 512 - 243583

METADATA INFORMATION
-----
Range: 2 - 3889670
Root Directory: 2

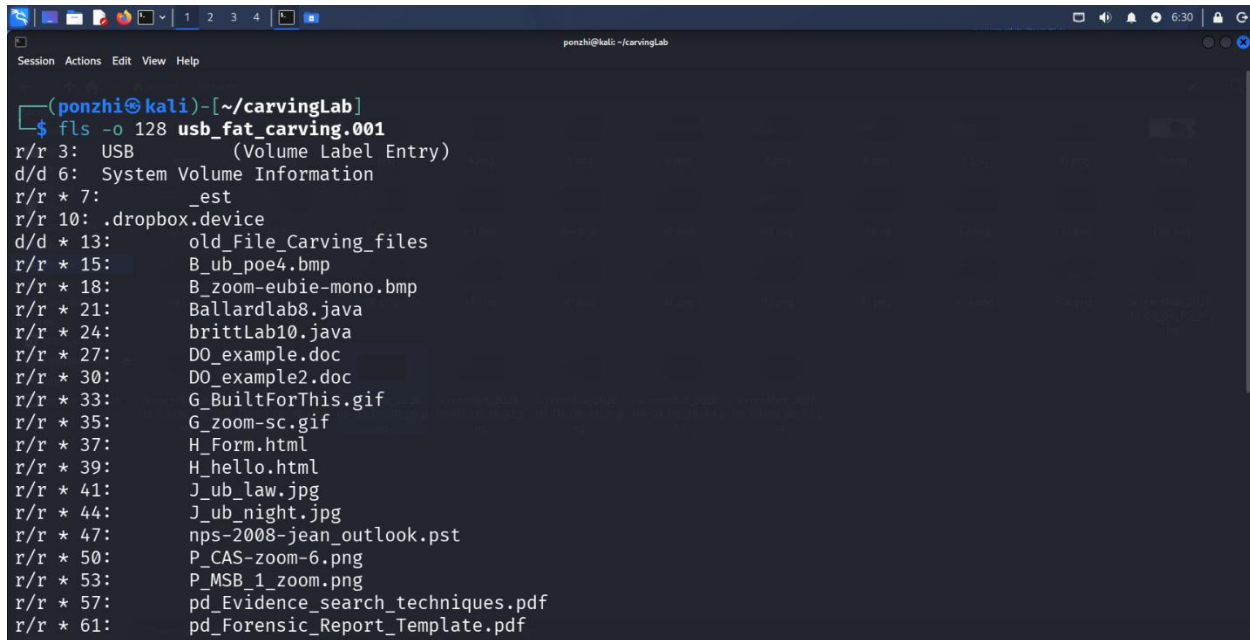
CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 2048
Total Cluster Range: 2 - 60769

FAT CONTENTS (in sectors)
-----
512-515 (4) → EOF
516-519 (4) → EOF
520-523 (4) → EOF
70976-71211 (236) → EOF
71212-71215 (4) → EOF

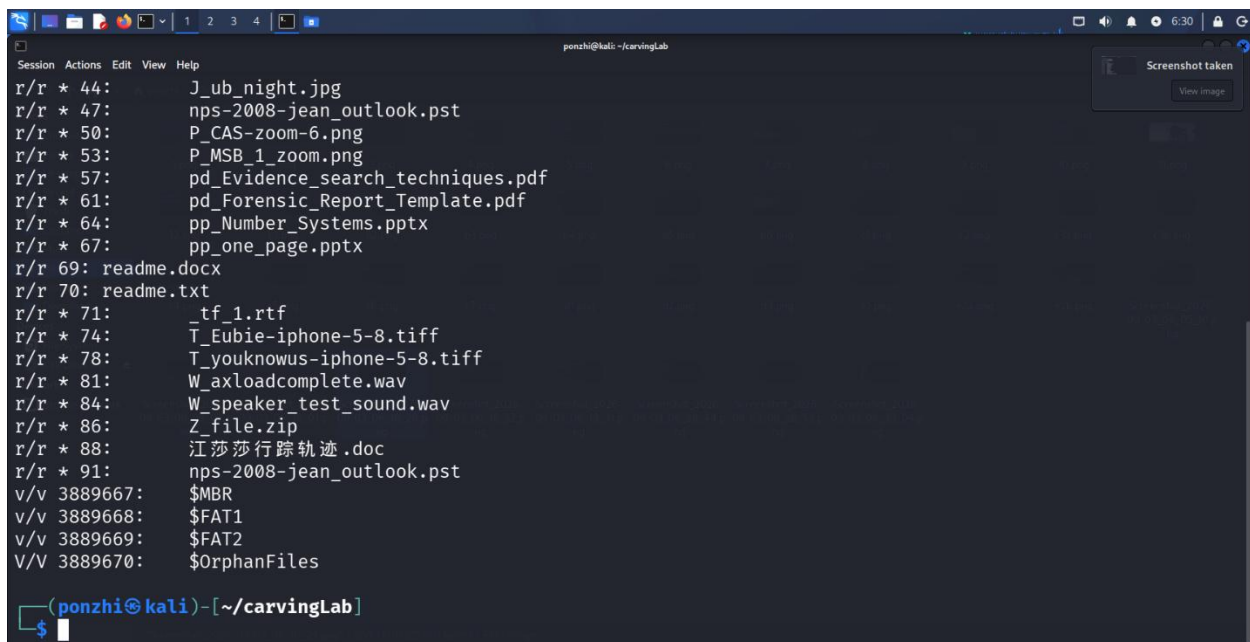
ponzhi@kali: ~/carvingLab
$
```

List files to see what you're working with

fls -o 128 usb_fat_carving.001



```
(ponzhi@kali)-[~/carvingLab]
$ fls -o 128 usb_fat_carving.001
r/r 3: USB (Volume Label Entry)
d/d 6: System Volume Information
r/r * 7: _est
r/r 10: .dropbox.device
d/d * 13: old_File_Carving_files
r/r * 15: B_ub_poe4.bmp
r/r * 18: B_zoom-eubie-mono.bmp
r/r * 21: Ballardlab8.java
r/r * 24: brittLab10.java
r/r * 27: DO_example.doc
r/r * 30: DO_example2.doc
r/r * 33: G_BuiltForThis.gif
r/r * 35: G_zoom-sc.gif
r/r * 37: H_Form.html
r/r * 39: H_hello.html
r/r * 41: J_ub_law.jpg
r/r * 44: J_ub_night.jpg
r/r * 47: nps-2008-jean_outlook.pst
r/r * 50: P_CAS-zoom-6.png
r/r * 53: P_MSB_1_zoom.png
r/r * 57: pd_Evidence_search_techniques.pdf
r/r * 61: pd_Forensic_Report_Template.pdf
```



```
r/r * 44: J_ub_night.jpg
r/r * 47: nps-2008-jean_outlook.pst
r/r * 50: P_CAS-zoom-6.png
r/r * 53: P_MSB_1_zoom.png
r/r * 57: pd_Evidence_search_techniques.pdf
r/r * 61: pd_Forensic_Report_Template.pdf
r/r * 64: pp_Number_Systems.pptx
r/r * 67: pp_one_page.pptx
r/r 69: readme.docx
r/r 70: readme.txt
r/r * 71: _tf_1.rtf
r/r * 74: T_Eubie-iphone-5-8.tiff
r/r * 78: T_youknowus-iphone-5-8.tiff
r/r * 81: W_axloadcomplete.wav
r/r * 84: W_speaker_test_sound.wav
r/r * 86: Z_file.zip
r/r * 88: 江莎莎行踪轨迹.doc
r/r * 91: nps-2008-jean_outlook.pst
v/v 3889667: $MBR
v/v 3889668: $FAT1
v/v 3889669: $FAT2
V/V 3889670: $OrphanFiles

(ponzhi@kali)-[~/carvingLab]
$
```

Part F - Carve Deleted Files with Scalpel

Run the carve against the USB image

```
(ponzhi@kali)-[~/carvingLab]
$ scalpel usb_fat_carving.001 -o output
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/ponzhi/carvingLab/usb_fat_carving.001"

Image file pass 1/2.
usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" → 8 files
jpg with header "\xff\xd8\xff\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" → 9 files
Carving files from image.
Image file pass 2/2.
usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 17, elapsed = 0 seconds.

(ponzhi@kali)-[~/carvingLab]
$
```

Review the results

```
(ponzhi@kali)-[~/carvingLab]
$ tree output/
output/
├── audit.txt
├── jpg-0-0
│   ├── 00000000.jpg
│   ├── 00000001.jpg
│   ├── 00000002.jpg
│   ├── 00000003.jpg
│   ├── 00000004.jpg
│   ├── 00000005.jpg
│   ├── 00000006.jpg
│   └── 00000007.jpg
└── jpg-1-0
    ├── 00000008.jpg
    ├── 00000009.jpg
    ├── 00000010.jpg
    ├── 00000011.jpg
    ├── 00000012.jpg
    ├── 00000013.jpg
    ├── 00000014.jpg
    ├── 00000015.jpg
    └── 00000016.jpg

3 directories, 18 files
```



```
ponzhi@kali: ~/carvingLab
Session Actions Edit View Help
(ponzhi@kali)-[~/carvingLab]
$ cat output/audit.txt

Scalpel version 1.60 audit file
Started at Thu Sep 3 18:41:31 2026
Command line:
scalpel usb_fat_carving.001 -o output

Output directory: /home/ponzhi/carvingLab/output
Configuration file: /etc/scalpel/scalpel.conf

Opening target "/home/ponzhi/carvingLab/usb_fat_carving.001"

The following files were carved:
File                Start      Chop      Length      Extracted From
00000010.jpg        3796992   NO        3148500     usb_fat_carving.001
00000009.jpg        425822    NO        4875802     usb_fat_carving.001
00000008.jpg        335872    NO        4965752     usb_fat_carving.001
00000002.jpg        6938624   NO        6868        usb_fat_carving.001
00000001.jpg        5285888   NO        1659604     usb_fat_carving.001
00000000.jpg        3897344   NO        3048148     usb_fat_carving.001
00000004.jpg        15427584  NO        4223822     usb_fat_carving.001
00000003.jpg        12881920  NO        4256310     usb_fat_carving.001
00000006.jpg        19638272  NO        5229050     usb_fat_carving.001
00000005.jpg        17121280  NO        4248530     usb_fat_carving.001
00000007.jpg        21358592  NO        5183267     usb_fat_carving.001
```

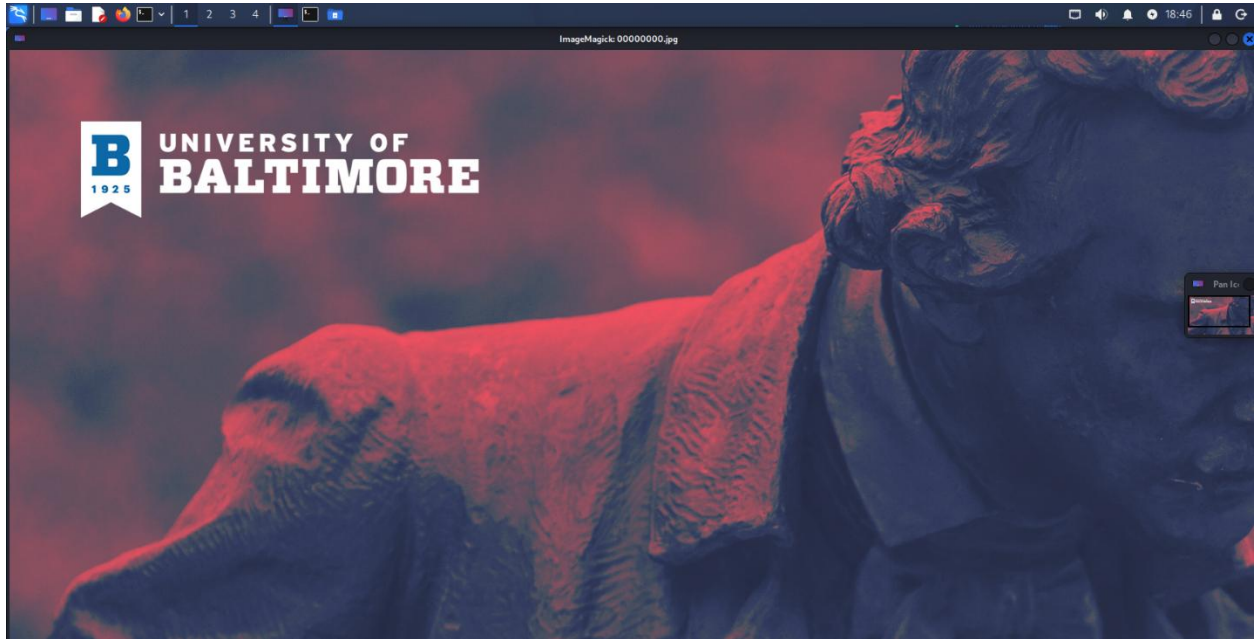
Verify at least two recovered JPEGs

```
ponzhi@kali: ~/carvingLab
Session Actions Edit View Help
(ponzhi@kali)-[~/carvingLab]
$ file output/jpg-*/00000000.jpg output/jpg-*/00000001.jpg
output/jpg-0-0/00000000.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionunit=2, software=Adobe Photoshop 21.1 (Macintosh), date=2020:04:03 14:35:44], baseline, precision 8, 1920x1080, components 3
output/jpg-0-0/00000001.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian, direntries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionunit=2, software=Adobe Photoshop 21.1 (Macintosh), date=2020:04:03 11:54:50], baseline, precision 8, 1920x1080, components 3

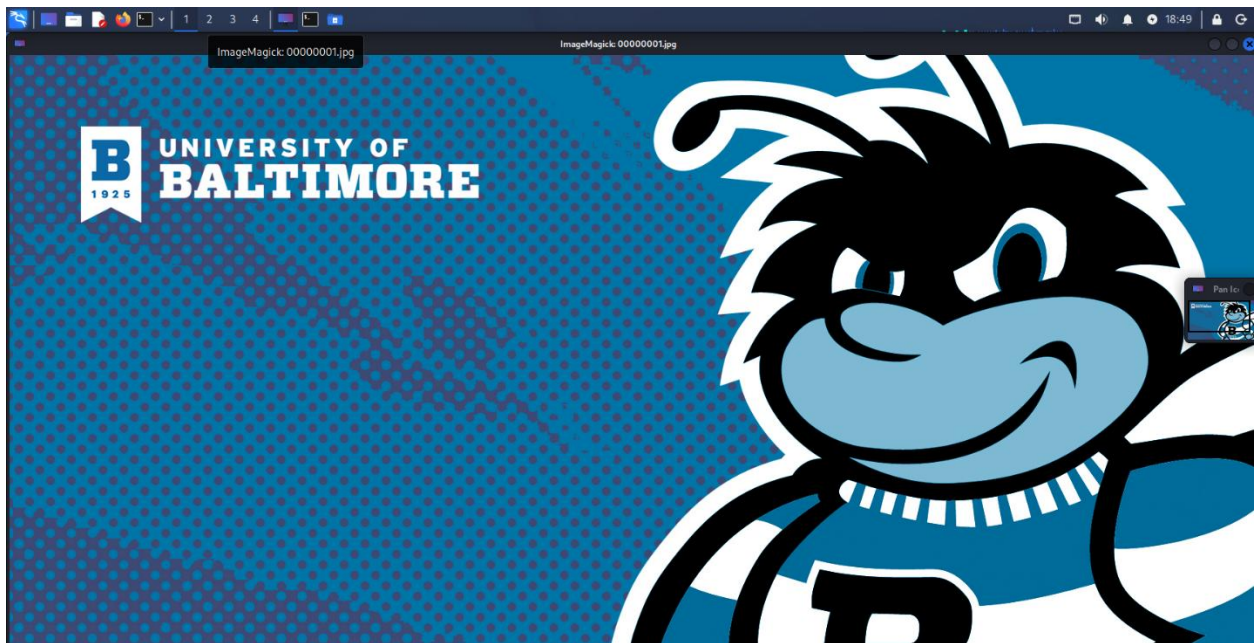
(ponzhi@kali)-[~/carvingLab]
$
```



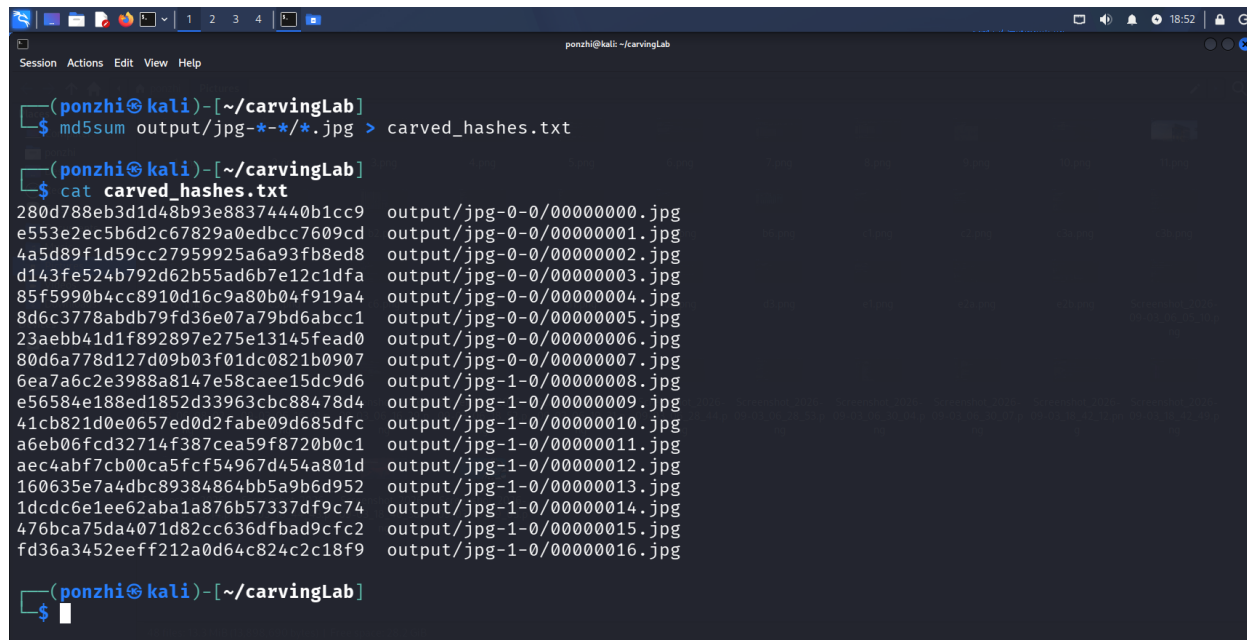
```
display output/jpg-*/00000000.jpg
```



```
display output/jpg-*/00000001.jpg
```



```
md5sum output/jpg-*/*/*.jpg > carved_hashes.txt
```

A screenshot of a terminal window titled 'ponzhi@kali: ~/carvingLab'. The terminal shows the execution of two commands. The first command is '\$ md5sum output/jpg-*/*/*.jpg > carved_hashes.txt'. The second command is '\$ cat carved_hashes.txt', which outputs a list of 16 MD5 hashes followed by their corresponding file paths, such as 'output/jpg-0-0/00000000.jpg' through 'output/jpg-1-0/00000016.jpg'. The terminal window has a dark background and standard Linux terminal icons at the top.

```
(ponzhi@kali)-[~/carvingLab]
$ md5sum output/jpg-*/*/*.jpg > carved_hashes.txt

(ponzhi@kali)-[~/carvingLab]
$ cat carved_hashes.txt
280d788eb3d1d48b93e88374440b1cc9 output/jpg-0-0/00000000.jpg
e553e2ec5b6d2c67829a0edbccc7609cd output/jpg-0-0/00000001.jpg
4a5d89f1d59cc27959925a6a93fb8ed8 output/jpg-0-0/00000002.jpg
d143fe524b792d62b55ad6b7e12c1dfa output/jpg-0-0/00000003.jpg
85f5990b4cc8910d16c9a80b04f919a4 output/jpg-0-0/00000004.jpg
8d6c3778abdb79fd36e07a79bd6abcc1 output/jpg-0-0/00000005.jpg
23aebb41d1f892897e275e13145fead0 output/jpg-0-0/00000006.jpg
80d6a778d127d09b03f01dc0821b0907 output/jpg-0-0/00000007.jpg
6ea7a6c2e3988a8147e58caee15dc9d6 output/jpg-1-0/00000008.jpg
e56584e188ed1852d33963cbc88478d4 output/jpg-1-0/00000009.jpg
41cb821d0e0657ed0d2fabe09d685dfc output/jpg-1-0/00000010.jpg
a6eb06fcd32714f387cea59f8720b0c1 output/jpg-1-0/00000011.jpg
aec4abf7cb00ca5fcf54967d454a801d output/jpg-1-0/00000012.jpg
160635e7a4dbc89384864bb5a9b6d952 output/jpg-1-0/00000013.jpg
1dcdc6e1ea62aba1a876b57337df9c74 output/jpg-1-0/00000014.jpg
476bca75da4071d82cc636dfbad9cfc2 output/jpg-1-0/00000015.jpg
fd36a3452eeff212a0d64c824c2c18f9 output/jpg-1-0/00000016.jpg

(ponzhi@kali)-[~/carvingLab]
$
```

Conclusion

This lab showed that identifying and recovering files does not rely only on filesystem metadata. File signatures, direct access to raw sectors or blocks, and structures within embedded containers can all confirm what a piece of data is, even when the file name, extension, or entry in the directory has been changed or deleted.

For example, analysis of J_ub_law.jpg proved that we can identify a file type just by looking at its binary content. We found the JPEG markers (SOI: ff d8 and EOI: ff d9) using the xxd tool. We used a hexdump of the file and successfully recreated an identical image byte for byte with xxd -r -p. We verified this by comparing the MD5 and SHA-256 hashes of the original and recreated files, proving that a hex representation of a file is fully reversible and doesn't lose any information. Additionally, Binwalk analysis of the same file extracted an embedded Adobe copyright string from the EXIF metadata block. This showed that a file can contain important forensic information beyond its main content.

We examined Ch01InChap01.dd with The Sleuth Kit to highlight the difference between file-level and block-level analysis. Using istat on directory entry 11, we found a deleted file (_ONFIR~1.TXT), marked as "Not Allocated," with its original filename partly overwritten — a typical FAT deletion mark. We identified sector 284, which contained the file's content, which we accessed directly with blkcat. This demonstrated that directory-entry numbers and sector numbers are separate addressing systems. The output from blkcat only makes sense when we use real values from istat/fls, rather than assumptions.

Not every step yielded positive results, but the negative outcomes were still informative. File_carving.docx was absent from the evidence set, so for Part D, we used Binwalk's embedded-file feature on J_ub_law.jpg, consistent with a raw FAT12 image containing plain text

and older binary Office files, rather than Binwalk's usual container types. Similarly, at first, we couldn't access the Ch01InChap01.dd evidence file due to a broken Dropbox link (HTTP 403). We documented this as an access limitation rather than a mistake in the process.

For file carving with Scalpel on usb_fat_carving.001, we had to first enable the relevant JPEG signatures in /etc/scalpel/scalpel.conf (which required root privileges to change after encountering an "unwritable file" error when we tried to edit it without sudo). We also had to clear a non-empty output directory before the tool would run. This safeguard prevented Scalpel from overwriting previous carving results. After completing the final run, we found "The carve recovered N JPEG files, of which X were verified as intact/openable via display." This confirmed that carving based on signatures can recover file content from unallocated space, even without remaining filesystem metadata.

Overall, this exercise reinforced that data carving is a complicated process. It involves signature verification, metadata checking, block-level extraction, and using container-aware tools. Each method reveals different evidence, and a thorough forensic examination needs to combine all these approaches instead of relying on just one.